

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUBJECT: Foundations of Computer Security

■ CS 6301 ■ Duration 120 minutes ■ Full Marks 30 ■ Date: 23-September-2022■

Answer ALL Questions

All parts of the question must be answered at one place.

Number within bracket indicate full mark for that question.

1. (a) Find a single value x that simultaneously solves the two congruences [2]
 $x \equiv 13 \pmod{71}$ and $x \equiv 41 \pmod{97}$
- (b) Let $p = 541$ and let the key k in Affine cipher be $(37, 71)$. Encrypt the message $m = 204$ and decrypt the cipher text $c = 431$. [3]
- (c) Find the multiplicative inverse of 8 in Z_{61}^* using extended Euclidean algorithm. [1]
- (d) Consider the Hill Cipher defined by $e_k(m) \equiv k_1 \cdot m + k_2 \pmod{p}$ and $d_k(m) \equiv k_1^{-1} \cdot (c - k_2) \pmod{p}$ where m, c, k_2 are column vectors of dimension n , and k_1 is an $n \times n$ matrix. We use the vector Hill cipher with $p = 7$ and the key $k_1 = \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$ and $k_2 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$. [4]
 - i. Encrypt the message $m = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.
 - ii. Decrypt the message $c = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$.
2. (a) Design a polybious key matrix for play fair cipher using the keyword "ODISHA" and encrypt the message defend the east wall of the castle. [3]
- (b) Use extended Euclidean algorithm to find the inverse of $x^3 + x + 1$ in $GF(2^{10})$ using the modulus $x^{10} + x^3 + 1$. [2]
- (c) What are the various components used in modern symmetric key encipherment? Discuss each with neat diagram. [2]
- (d) Discuss the working procedure of Enigma machine with suitable diagram. [3]
3. (a) Find a particular and general solution to the following Diophantine equation. [2]
 $28x + 42y = 154$
- (b) Encrypt the first 8 characters of your name using a Vigenere cipher and the key is **PILL**. Start with your first name, and if it is less than 8 characters continue with your second name, ignore spaces. [2]
- (c) The encryption key in the transposition cipher is $(6, 4, 3, 1, 5, 2)$. Find the decryption key. [2]
- (d) Assume I designed my own cipher to encrypt exam answers. The cipher uses a 40/50 bit key. I always create the answers 10 days before the exam and store them on my office computer, however on the same day that I created the exam answers a student obtained access to my office computer and took a copy of the answers. The student has free access to a computer that can decrypt the answers at a rate of 200×10^9 attempts per day (as the student wants to cheat on the exam, he doesn't know about any limitations in the encryption algorithm). Is this system used for encrypting exams computationally secure? Explain your answer, showing any calculations where necessary. [4]
(Note that $2^{10} \approx 10^3$, $2^{20} \approx 10^6$, $2^{30} \approx 10^9$, and so on...)

*****Wish you all the best*****